Question-1

Find the poisson distrubution formula by using limiting case of the Binomial distribution.

Binomial distribution approaches to Poisson distribution as $n \to \infty$.

For Binomial distribution,

$$P(X=r) = {}^nC_r\, p^r q^{n-r} \qquad r = 0, 1, 2, \cdots ; n.$$

where 'p' is the probability of successes and 'q' is the probability of failures. There are 'r' successes and 'n-r' failures in 'n' trials.

Now,

$$P(X=r) = {}^nC_r\, p^r q^{n-r} \qquad r = 0, 1, 2, \cdots, n.$$

$$= \frac{n(n-1)(n-2)(n-3)\cdots(n-r+1)}{r!} p^r (1-p)^{n-r}$$

$$= \frac{np(np-p)(np-2p)(np-3p)\cdots[np-rp+p]}{r!} \cdot \frac{(1-p)^n}{(1-p)^r}$$

$\because$ We know, $m = np$

$$= \frac{m(m-p)(m-2p)(m-3p)\cdots(m-rp+p)}{r!} \cdot \left(\frac{(1-p)^n}{(1-p)^r}\right)$$

$\because$ $m = np \Rightarrow p = \frac{m}{n}$

$$= \frac{m\left(m-\frac{m}{n}\right)\left(m-\frac{2m}{n}\right)\left(m-\frac{3m}{n}\right)\cdots\left(m-(r-1)\frac{m}{n}\right)}{r!} \cdot \frac{\left(1-\frac{m}{n}\right)^{n}}{\left(1-\frac{m}{n}\right)^{r}}$$

Taking limit as $n \to \infty$,

We know, $\lim_{n\to\infty}\left(1-\frac{m}{n}\right)^{n} = e^{-m}$

& $\lim_{n\to\infty}\left(1-\frac{m}{n}\right)^{r} = 1$

$$P(X=r) = \frac{m^{r}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\left(1-\frac{3}{n}\right)\cdots\left(1-\frac{(r-1)}{n}\right)\cdot\left(1-\frac{m}{n}\right)^{n}}{r!\left(1-\frac{m}{n}\right)^{r}}$$

$$P(X=r) = \frac{m^{r}\cdot e^{-m}}{r!}$$

$$\therefore P(X=r) = \frac{e^{-m}\cdot m^{r}}{r!} \qquad r = 0, 1, 2, \cdots, \infty$$

which is the Poisson distribution probability function, and 'm' is called parameter of the Poisson distribution.

Question-2

By using Binomial Expression, find the formula for Binomial Distribution.

⇒ Consider an experiment consisting of n independent trials, known as Bernoulli trials.

Let p be the probability of success (S) in a single trial.
Let q be the probability of failure (F) in a single trial.
Since, success and failure are the only two outcomes,
$p+q=1$.
The trials are independent, meaning the probability p remains constant for every trial.
Let the random variable X denote the number of successes in n trials.

We want to find the probability of obtaining exactly r successes and $(n-r)$ failures. Suppose the successes occur in a specific order.

$$P(\underbrace{S \cdot S \cdot S \cdots S}_{r \text{ factors}} \cdot \underbrace{F \cdot F \cdot F \cdots F}_{n-r \text{ factors}})$$

Since the trials are independent, the probability of this specific sequence is the product of their individual probabilities:

$$P(\text{Specific Sequence}) = (p \cdot p \cdots \text{up to } r \text{ times}) \times (q \cdot q \cdots \text{upto } n-r \text{ times})$$

$$= p^r q^{n-r}$$

In n trials, r successes can occur in many different positions. The number of ways to choose r positions for successes out of n total trials is given by the combination formula:

Number of ways $= {}^nC_r$

The total probability of r successes is the number of possible sequences multiplied by the probability of any one sequence:

$$P(X=r) = {}^nC_r\, p^r q^{n-r}$$

where $r = 0, 1, 2, \dots n$

The probabilities $P(X=r)$ for $r = 0, 1, 2, \dots, n$ are the successive terms in the binomial expansion of $(q+p)^n$.

$$(q+p)^n = {}^nC_0\, q^n + {}^nC_1 \cdot q^{n-1} p + {}^nC_2 \cdot q^{n-2} p^2 + \dots + {}^nC_n\, p^n$$

This is why the distribution is called Binomial Distribution.

Question-3 Find the probability density function of Normal Distribution.

$\Rightarrow$ The probability density function of the normal distribution is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Area under the curve, $A = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$

Let $\frac{x-\mu}{\sigma} = z$ so $dx = \sigma dz$

$$A = \frac{\cancel{\sigma}}{\cancel{\sigma}\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2} dz = \frac{\cancel{2}\sqrt{2}}{\cancel{\sqrt{2}}\sqrt{\pi}} \int_{0}^{\infty} e^{-z^2/2} dz$$

Since $e^{-z^2/2}$ is an even function.

Let $z^2/2 = t \Rightarrow \frac{\cancel{2}z\,dz}{\cancel{2}} = dt$

and $z = \sqrt{2t}$

$\therefore \quad dz = \frac{dt}{z} = \frac{dt}{\sqrt{2t}}$

$$A = \frac{\sqrt{2}}{\sqrt{\pi}} \int_0^\infty e^{-t} \cdot \frac{dt}{\sqrt{2t}} = \frac{\cancel{\sqrt{2}}}{\sqrt{\pi}\cancel{\sqrt{2}}} \int_0^\infty e^{-t} t^{-1/2} dt$$

$$A = \frac{1}{\sqrt{\pi}} \int_0^\infty e^{-t} t^{1/2 - 1} dt$$

We know, $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$

$\therefore \quad A = \frac{1}{\sqrt{\pi}} \cdot \Gamma\left(\frac{1}{2}\right) \qquad \because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

$$A = \frac{1}{\cancel{\sqrt{\pi}}} \cdot \cancel{\sqrt{\pi}} = 1$$

$\therefore$ A = Area under the Normal curve $= 1$

Question-4

By using Binomial Distribution, find the formula of Normal Distribution

Let the binomial distribution be $N(q+p)^n$. Given, $p = q = \frac{1}{2}$, the distribution is perfectly symmetrical.

Let n be an even integer, $n = 2k$

As $n \to \infty$, $k \to \infty$.

The frequency of r successes is :

$$f(r) = N \cdot {}^{2k}C_r \cdot \left(\frac{1}{2}\right)^{2k}$$

To find where the frequency is maximum, we compare $f(r)$ and $f(r+1)$:

$$\frac{f(r+1)}{f(r)} = \frac{{}^{2k}C_{r+1}}{{}^{2k}C_{r}} = \frac{2k-(r+1)+1}{r+1} = \frac{2k-r}{r+1}$$

The frequency $f(r)$ will be maximum when the ratio is approximately 1. (where the "growth" stops and "decay" begin). Solving $2k-r \approx r+1$ gives $r \approx k$.

Thus the maximum frequency occurs at $r=k$, which we denote as y_0.

$$y_0 = N \cdot {}^{2k}C_k \left(\frac{1}{2}\right)^{2k} = N \cdot \frac{(2k)!}{k!\,k!} \left(\frac{1}{2}\right)^{2k}$$

Let y_x be the frequency of $(k+x)$ successes, where x is a small deviation from mean k.

$$y_x = N \cdot \frac{(2k)!}{(k+x)!\,(k-x)!} \left(\frac{1}{2}\right)^{2k}$$

Taking the ratio of y_x to y_0:

$$\frac{y_x}{y_0} = \frac{\dfrac{\cancel{N} \cdot \cancel{(2k)!}}{(k+x)!\,(k-x)!} \cancel{\left(\frac{1}{2}\right)^{2k}}}{\dfrac{\cancel{N} \cdot \cancel{(2k)!}}{k!\,k!} \cancel{\left(\frac{1}{2}\right)^{2k}}}$$

$$\frac{y_x}{y_0} = \frac{k!\,k!}{(k+x)(k-x)!} = \frac{(k!)\,k \cdot (k-1)(k-2)\cdots(k-(x-1))\cancel{(k-x)!}}{(k+x)!\,\cancel{(k-x)!}}$$

$$\frac{y_x}{y_0} = \frac{\cancel{k!}\; k \cdot (k-1)(k-2)\cdots(k-(x+1))}{(k+x)(x+(x-1))(k+(x-2))\cdots \cancel{k!}}$$

$$\frac{y_x}{y_0} = \frac{\left(1-\frac{1}{k}\right)\left(1-\frac{2}{k}\right)\cdots\left(1-\frac{x-1}{k}\right)}{\left(1+\frac{1}{k}\right)\left(1+\frac{2}{k}\right)\cdots\left(1+\frac{x}{k}\right)}$$

Taking natural logs on both sides:

$$\ln\left(\frac{y_x}{y_0}\right) = \ln\left(1-\frac{1}{k}\right) + \ln\left(1-\frac{2}{k}\right) + \ln\left(1-\frac{3}{k}\right) + \cdots + \ln\left(1-\frac{x-1}{k}\right)$$
$$-\left[\ln\left(1+\frac{1}{k}\right) + \ln\left(1+\frac{2}{k}\right) + \ln\left(1+\frac{3}{k}\right) + \cdots + \ln\left(1+\frac{x}{k}\right)\right]$$

$$= \sum_{i=1}^{x-1} \ln\left(1-\frac{i}{k}\right) - \sum_{i=1}^{x} \ln\left(1+\frac{i}{k}\right)$$

Using the expansion $\ln(1\pm\delta) \approx \pm\delta$
(neglecting the higher powers of $\frac{1}{k}$ since $k \to \infty$)

$$\ln \frac{y_x}{y_0} \approx -\sum_{i=1}^{x-1}\left(\frac{i}{k}\right) - \sum_{i=1}^{x}\left(\frac{i}{k}\right)$$

$$\approx -\frac{1}{k}(1+2+\cdots+(x-1)) - \frac{1}{k}(1+2+\cdots+x)$$

$$\approx -\frac{1}{k}\left[\frac{(x-1)x}{2} + \frac{x(x+1)}{2}\right]$$

$$\approx -\frac{1}{k}\left[\frac{x(x+\cancel{1}+x-\cancel{1})}{2}\right] = -\frac{2x^2}{2k} = -\frac{x^2}{k}$$

Taking the antilogs

$$y_x = y_0 \cdot e^{-x^2/k}$$

For a Binomial Distribution, variance

$$\sigma^2 = npq$$

Substituting $n = 2k$ and $p = q = \frac{1}{2}$

$$\sigma^2 = (2k)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{k}{2} \Rightarrow k = 2\sigma^2$$

Substituting k back into the equation:

$$y_x = y_0 \cdot e^{-\frac{x^2}{2\sigma^2}}$$

This is the equation of the Normal curve, proving that the Binomial Distribution tends towards the Normal Distribution as n becomes large.

Question-5

Find mean of the Normal Distribution.

The normal distribution is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \qquad -\infty \le X \le \infty$$

$$\text{Mean} = E(X) = \int_{-\infty}^{\infty} x\, f(x)\,dx = \int_{-\infty}^{\infty} x \cdot \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

Put $\frac{x-\mu}{\sigma} = z$, then $dx = \sigma dz$ and $x = \mu + \sigma z$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + \sigma z) e^{-\frac{1}{2}z^2} \sigma\, dz$$

$$= \frac{\mu}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} dz + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z \cdot e^{-\frac{1}{2}z^2} dz$$

Here, $\because$ $z e^{-\frac{1}{2}z^2}$ is an odd function and

$e^{-\frac{1}{2}z^2}$ is an even function

$$= \frac{\mu}{\sqrt{2\pi}} \cdot 2\int_{0}^{\infty} e^{-\frac{1}{2}z^2} dz + 0$$

Now, $\because$ $\int_{0}^{\infty} e^{-\frac{1}{2}z^2} dx = \sqrt{\frac{\pi}{2}}$

$$= \frac{\mu}{\cancel{\sqrt{2\pi}}} \cdot \frac{\cancel{2} \cdot \cancel{\sqrt{\pi}}}{\cancel{\sqrt{2}}} = \mu$$

$\therefore$ $E(X) = \mu$

$\therefore$ The mean of the Normal Distribution is 'μ'.

Question-6

Find variance of the Normal Distribution

Since $\quad Var(X) = E(X^2) - [E(X)]^2 = E(X-\mu)^2$

Now $\quad E(X-\mu)^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx = \int_{-\infty}^{\infty} (x-\mu)^2 \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$

Put $\frac{x-\mu}{\sigma} = z$ then $dx = \sigma\,dz$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \sigma^2 z^2 e^{-\frac{1}{2}z^2} \sigma\,dz = \frac{2\sigma^2}{\sqrt{2\pi}} \int_{0}^{\infty} z^2 \cdot e^{-\frac{1}{2}z^2} dz$$

$\because z^2 \cdot e^{-\frac{1}{2}z^2}$ is an even function

Let $\frac{z^2}{2} = t$, then $z\,dz = dt$ and $z = \sqrt{2t}$

$$= \frac{2\sigma^2}{\sqrt{2\pi}} \int_{0}^{\infty} 2t \cdot e^{-t} \cdot \frac{dt}{\sqrt{2t}} = \frac{2\sigma^2}{\sqrt{2}\cdot\sqrt{\pi}} \cdot \int_{0}^{\infty} \sqrt{2t} \cdot e^{-t} \cdot dt$$

$$= \frac{2\sigma^2}{\cancel{\sqrt{2}}\sqrt{\pi}} \cdot \cancel{\sqrt{2}} \int_{0}^{\infty} (t)^{1/2} \cdot e^{-t} \cdot dt = \frac{2\sigma^2}{\sqrt{\pi}} \int_{0}^{\infty} e^{-t} \cdot t^{3/2-1} dt$$

We know, by Gamma function, $\int_{0}^{\infty} e^{-t} \cdot t^{3/2-1} dt = \Gamma\left(\frac{3}{2}\right)$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(\frac{3}{2}\right) = \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{\cancel{2}\sigma^2 \cancel{\sqrt{\pi}}}{\cancel{\sqrt{\pi}} \cdot \cancel{2}} = \sigma^2$$

$\because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

$\therefore$ Variance of Normal Distribution is σ^2

Question-7

Fit a binomial distribution to the frequency data:

x:	0	1	2	3	4	5	6	7
f:	21	34	2	18	16	5	0	0

x	f	fx
0	21	0
1	34	34
2	2	4
3	18	54
4	5	64
5	5	25
6	0	0
7	0	0
	$\Sigma f = 96$	$\Sigma fx = 181$

Mean of observations $= \dfrac{\Sigma fx}{\Sigma f} = \dfrac{181}{96} = 1.885$

$$\Rightarrow \quad np = 1.885 \Rightarrow 7 \times p = 1.885$$

$$p = \frac{1.885}{7} = 0.2693$$

$$q = 1 - p = 1 - 0.2693 = 0.731$$

Also, $N = 96$

Hence, the binomial distribution is $= N(q+p)^n$

$$= 96(0.269 + 0.731)^7$$

Question-8

A manufacturer knows that the condensors he makes contain on an average 1% of defectives. He packs them in boxes of 100. What is the probability that a box picked at random will contain 4 or more faulty condensors?

Solⁿ: Given, $p = 0.01$, $n = 100$

We know, mean $m = np$

$$= 100 \times 0.01 = 1$$

By Poisson distribution, $P(r) = \dfrac{e^{-m} \cdot m^{r}}{r!}$

$P(4 \text{ or more faulty condensors}) = P(4) + P(5) + \cdots + P(100)$

$$= 1 - [P(0) + P(1) + P(2) + P(3)]$$

$$= 1 - \left[\frac{e^{-1}}{0!} + \frac{e^{-1}}{1!} + \frac{e^{-1}}{2!} + \frac{e^{-1}}{3!}\right]$$

$$= 1 - e^{-1}\left[1 + 1 + \frac{1}{2} + \frac{1}{6}\right] = 1 - \frac{1}{e}\left[\frac{6+6+3+1}{6}\right] = 1 - \frac{1}{e}\left[\frac{8}{3}\right]$$

$$= 1 - \frac{8}{3e} = 1 - 0.981 = 0.019 \text{ Ans.}$$

Question-9

Find the fourth moment.

We know, the formula for r^{th} moment is given by

$$\mu_r = \frac{\sum_{i=1}^{n} f_i (x_i - \bar{x})^r}{N}$$

where $N = \sum f_i$

Expanding $(x_i - \bar{x})^4 = x_i^4 - 4x_i^3\bar{x} + 6x_i^2\bar{x}^2 - 4x_i\bar{x}^3 + \bar{x}^4$

Multiply by f_i and sum:

$$\mu_4 = \frac{\sum f_i x_i^4 - 4\bar{x}\sum f_i x_i^3 + 6\bar{x}^2 \sum f_i x_i^2 - 4\bar{x}^3 \sum f_i x_i + \bar{x}^4 \sum f_i}{N}$$

Since $\sum f_i = N$ and $\sum f_i x_i = N\bar{x}$

After simplification,

$$\mu_4 = \frac{1}{N}\sum f_i x_i^4 - 4\bar{x}\frac{1}{N}\sum f_i x_i^3 + 6\bar{x}^2 \cdot \frac{1}{N}\sum f_i x_i^2 - 3\bar{x}^4$$

Question-10

Find the first four moments for the individual data.

x:	10	20	23	25	27
f:	3	5	9	12	16

Verify it with the poisson and binomial mean and variance.

Sol^n:

x	f	fx	$x-\bar{x}$	$f(x-\bar{x})$	$f(x-\bar{x})^2$	$f(x-\bar{x})^3$	$f(x-\bar{x})^4$
10	3	30	-13.75	-41.25	567.1875	-7798.83	107233.9
20	5	100	-3.75	-18.75	70.3125	-263.67	988.7695
23	9	207	0.75	-6.75	5.0625	-3.8	2.85
25	12	300	1.25	15	18.75	23.4375	29.3
27	16	432	3.25	52	169	549.25	1785.06
	$\sum f$	$\sum fx$	$\sum(x-\bar{x})$	$\sum f(x-\bar{x})$	$\sum f(x-\bar{x})^2$	$\sum f(x-\bar{x})^3$	$\sum f(x-\bar{x})^4$
	45	1069	-13.78	0	830.3125	-750745	110206.54

We know, mean, $\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{1069}{45} = 23.75$

$\therefore \mu_1 = \frac{\sum f(x-\bar{x})}{N} = \frac{0}{45} = 0$ $\quad \therefore \mu_2 = \frac{\sum f(x-\bar{x})^2}{N} = \frac{830.3125}{45} = 18.45$

$\therefore \mu_3 = \frac{\sum f(x-\bar{x})^3}{N} = \frac{-7507.45}{45} = -166.83$, $\quad \therefore \mu_4 = \frac{110206.54}{45} = 2449.03$

Calculated mean $= 23.75$

Calculated variance $(\mu_2) = 18.45$

- For a Poisson Distribution, Mean = Variance.

  Since, $23.75 \neq 18.45$, it is not a poisson distribution.

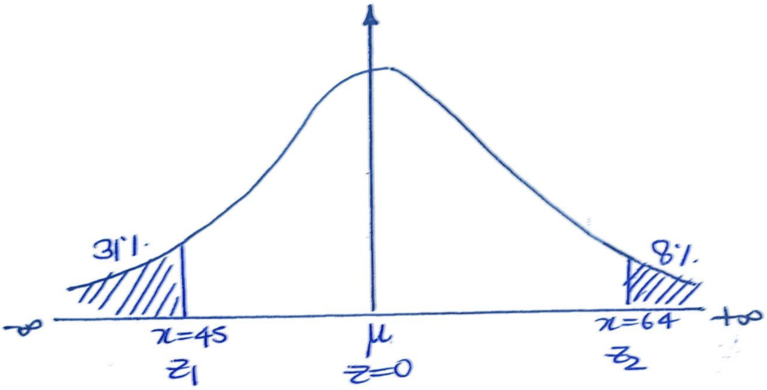
- For a Binomial Distribution, Mean > Variance.

  Since, $23.75 >$ Variance, the given data satisfies the property of Binomial distribution.

Question-11

In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and the standard deviation of the distribution.

Let μ be the mean and σ be the standard deviation



When $x=45$,

Let the standard normal variate be z_1.

$$z_1 = \frac{x-\mu}{\sigma} = \frac{45-\mu}{\sigma}$$

$$\Rightarrow \int_{-\infty}^{z_1} \phi(z)dz = 0.31 \quad \Rightarrow \int_{-\infty}^{0} \phi(z)dz - \int_{z_1}^{0} \phi(z)dz = 0.31$$

$$\Rightarrow 0.5 - \int_{z_1}^{0} \phi(z)dz = 0.31 \Rightarrow \int_{z_1}^{0} \phi(z)\,dz = 0.19$$

From table, we get $z_1 = -0.5$

$$\therefore \quad -0.5 = \frac{45-\mu}{\sigma} \Rightarrow -0.5\,\sigma = 45-\mu$$

$$\Rightarrow \mu - 0.5\sigma = 45 \quad \text{--- i)}$$

When $x=64$,

Let the standard normal variate be z_2,

$\Rightarrow z_2 = \frac{x-\mu}{\sigma} = \frac{64-\mu}{\sigma}$ $\quad \int_{z_2}^{\infty} \phi(z)dz = 0.08$

$\Rightarrow \int_0^{\infty} \phi(z)dz - \int_0^{z_2} \phi(z)dz = 0.08$

$\Rightarrow 0.5 - \int_0^{z_2} \phi(z)dz = 0.08 \Rightarrow \int_0^{z_2} \phi(z)dz = 0.5 - 0.08$

$\Rightarrow \int_0^{z_2} \phi(z)dz = 0.42$

From table, we get $z_2 = 1.4$

$\Rightarrow 1.4 = \frac{64-\mu}{\sigma} \Rightarrow 1.4\sigma = 64-\mu$

$\Rightarrow \mu + 1.4\sigma = 64$ - - - - ii)

From eq^n i) and eq^n ii),

$\mu - 0.5\sigma = 45$

$\mu + 1.4\sigma = 64$

Solving the above equations, we get

$\mu = 50$ and $\sigma = 10$

Question-12

If $z = ax + by$ and r is the correlation co-efficient between x and y, show that

$$\sigma_z^2 = a^2\sigma_x^2 + b^2\sigma_y^2 + 2abr\sigma_x\sigma_y$$

$\Rightarrow$ Since $z_i = ax_i + by_i$ --- eq^n i)

Taking mean, $\bar{z} = a\bar{x} + b\bar{y}$ --- eq^n ii)

Subtracting eq^n ii) from eq^n i)

$$z_i - \bar{z} = a(x_i - \bar{x}) + b(y_i - \bar{y}) \quad \text{--- eq}^n \text{ iii)}$$

This expresses deviation of z in terms of deviations of x and y.

We know by variance formula,

$$\sigma_z^2 = \frac{1}{n}\sum(z_i - \bar{z})^2$$

Substituting ~~eq^n iii) in~~ the above formula in eq^n iii)

$$\sigma_z^2 = \frac{1}{n}\sum\left[a(x_i - \bar{x}) + b(y_i - \bar{y})\right]^2$$

Expanding the square

$$\sigma_z^2 = \frac{1}{n}\sum\left[a^2(x_i - \bar{x})^2 + b^2(y_i - \bar{y})^2 + 2ab(x_i - \bar{x})(y_i - \bar{y})\right]$$

$$= \frac{1}{n}a^2\sum(x_i - \bar{x})^2 + \frac{b^2}{n}\sum(y_i - \bar{y})^2 + \frac{2ab}{n}\sum(x_i - \bar{x})(y_i - \bar{y})$$

We know, $\sigma_x^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$ and similarly

$$\sigma_y^2 = \frac{1}{n}\sum(y_i - \bar{y})^2$$

We also know, correlation coefficient

$$r = \frac{\frac{1}{n}\sum(x_i - \bar{x})(y_i - \bar{y})}{\sigma_x\sigma_y}$$

$\therefore \quad \sigma_Z^2 = a^2 \sigma_x^2 + b^2 \sigma_y^2 + 2abr\sigma_x \sigma_y$

Question-13

Write the formula of correlation using expectation values.

OR

State and prove Karl-Pearson's coefficient of correlation using expectation values.

⇒ The correlation coefficient ρ_{xy} is defined as the covariance of two variables divided by the product of their standard deviations :

$$\rho_{XY} = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

$$\rho_{XY} = \frac{E[(X - E[X])(Y - E[Y])]}{\sqrt{E[X^2] - (E[X])^2}\sqrt{E[Y^2] - (E[Y])^2}}$$

$$\rho_{XY} = \frac{E[XY - X\mu_Y - Y\mu_X + E[X]E[Y]]}{\sqrt{E[X^2] - (E[X])^2}\sqrt{E[Y^2] - (E[Y])^2}}$$

$$\rho_{XY} = \frac{E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X \mu_Y}{\sqrt{E[X^2] - (E[X])^2}\sqrt{E[Y^2] - (E[Y])^2}}$$

$$\rho_{XY} = \frac{E[XY] - \mu_X \mu_Y - \cancel{\mu_X \mu_Y} + \cancel{\mu_X \mu_Y}}{\sqrt{E[X^2] - (E[X])^2}\sqrt{E[Y^2] - (E[Y])^2}}$$

$$\rho_{XY} = \frac{E[XY] - E[X]E[Y]}{\sqrt{E[X^2] - (E[X])^2}\sqrt{E[Y^2] - (E[Y])^2}}$$

We know, $E[X] = \frac{\Sigma x}{n}$, $E[Y] = \frac{\Sigma y}{n}$

$E[XY] = \frac{\Sigma xy}{n}$, $E[X^2] = \frac{\Sigma x^2}{n}$, $E[Y^2] = \frac{\Sigma y^2}{n}$

Substituting these into formula :

$$r = \frac{\frac{\Sigma xy}{n} - \frac{\Sigma x}{n}\frac{\Sigma y}{n}}{\sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}\sqrt{\frac{\Sigma y^2}{n} - \left(\frac{\Sigma y}{n}\right)^2}} = \frac{\frac{n}{n}\cdot\frac{\Sigma xy}{n} - \frac{\Sigma x\,\Sigma y}{n\cdot n}}{\sqrt{\frac{\Sigma x^2}{n}\cdot\frac{n}{n} - \left(\frac{\Sigma x}{n}\right)^2}\sqrt{\frac{n\Sigma y^2}{n^2} - \left(\frac{\Sigma y}{n}\right)^2}}$$

$$r = \frac{\frac{1}{\cancel{n^2}}\left[n\Sigma xy - \Sigma x\Sigma y\right]}{\frac{1}{\cancel{n}}\sqrt{n\Sigma x^2 - (\Sigma x)^2}\cdot\frac{1}{\cancel{n}}\sqrt{n\Sigma y^2 - (\Sigma y)^2}}$$

$$r = \frac{n\Sigma xy - \Sigma x\Sigma y}{\sqrt{n\Sigma x^2 - (\Sigma x)^2}\sqrt{n\Sigma y^2 - (\Sigma y)^2}}$$

Question-14

Two variables x and y have zero means, the same variance σ^2 and zero correlation, show that

$$u = x\cos\alpha + y\sin\alpha \quad \text{and} \quad v = x\sin\alpha - y\cos\alpha$$

have the same variance σ^2 and zero correlation.

Sol^n: Given, $\bar{x} = 0$, $\bar{y} = 0$

$$\sigma_x^2 = \sigma_y^2 = \sigma^2$$

$$r(x,y) = 0 \Rightarrow Cov(x,y) = 0$$

and $u = x\cos\alpha + y\sin\alpha$

$v = x\sin\alpha - y\cos\alpha$

We know the formula :

$$Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2ab\,Cov(X,Y)$$

$$\therefore \quad \sigma_u^2 = \cos^2\alpha\,\sigma_x + \sin^2\alpha\,\sigma_y + 2\cos\alpha\sin\alpha \cdot Cov(x,y)$$

$$\sigma_u^2 = \sigma^2(\cos^2\alpha + \sin^2\alpha) + 0$$

$$\sigma_u^2 = \sigma^2$$

Now, $v = x\sin\alpha - y\cos\alpha$

$$\sigma_v^2 = \sin^2\alpha\,\sigma^2 + \cos^2\alpha \cdot \sigma^2$$
$$= \sigma^2(\sin^2\alpha + \cos^2\alpha) = \sigma^2$$

Now, $\mathrm{Cov}(u,v) = E[(u-\bar{u})(v-\bar{v})]$

$\because \bar{x} = \bar{y} = 0,\ \bar{u} = \bar{v} = 0$

So, $\mathrm{Cov}(u,v) = E[uv]$

$$= E[(x\cos\alpha + y\sin\alpha)(x\sin\alpha - y\cos\alpha)]$$
$$= E[x^2\sin\alpha\cos\alpha - xy\cos^2\alpha + xy\sin^2\alpha - y^2\sin\alpha\cos\alpha]$$
$$= E[(x^2-y^2)\sin\alpha\cos\alpha + xy(\sin^2\alpha - \cos^2\alpha)]$$
$$= \sin\alpha\cos\alpha[E(x^2-y^2)] + (\sin^2\alpha - \cos^2\alpha)[E(xy)]$$

Using given conditions, $E(x^2) = \sigma^2$, $E(y^2) = \sigma^2$

So, $E(x^2) - E(y^2) = 0$

And since covariance is zero : $E(xy) = 0$.

$\therefore \mathrm{Cov}(u,v) = 0 + 0 = 0$

$\therefore$ u and v have the same variance σ^2 but zero correlation.

Question-15

Find the angle between two lines of regression.

⇒ Equations to the lines of regression y on x is

$$y-\bar{y} = r\frac{\sigma_y}{\sigma_x}(x-\bar{x}) \quad \text{--- eq}^n \text{ i)}$$

and equation to the line of regression x on y is

$$x-\bar{x} = r\frac{\sigma_x}{\sigma_y}(y-\bar{y}) \quad \text{--- eq}^n \text{ ii)}$$

Let their slopes are m_1 and m_2.

$$\therefore \quad m_1 = r\frac{\sigma_y}{\sigma_x} \quad \text{and} \quad m_2 = r\frac{\sigma_x}{\sigma_y}$$

Rearranging eq^n ii), we get

$$y-\bar{y} = \frac{\sigma_y}{r\sigma_x}(x-\bar{x})$$

$$\therefore \quad m_2 = \frac{\sigma_y}{r\sigma_x}$$

We know, angle between two lines when their slope are m_1 and m_2 is

$$\therefore \quad \tan\theta = \pm\frac{m_2-m_1}{1+m_2 \cdot m_1} = \pm\frac{\frac{\sigma_y}{r\sigma_x} - r\cdot\frac{\sigma_y}{\sigma_x}}{1+\frac{\sigma_y^2}{\sigma_x^2}}$$

$$= \pm\frac{\left(\frac{\sigma_y}{\sigma_x}\right)\left(\frac{1-r^2}{r}\right)}{\frac{\sigma_x^2+\sigma_y^2}{\sigma_x^2}} = \pm\frac{1-r^2}{r}\cdot\frac{\sigma_y}{\cancel{\sigma_x}}\cdot\frac{\sigma_x^{\cancel{2}}}{\sigma_x^2+\sigma_y^2}$$

$$= \pm\frac{1-r^2}{r}\cdot\frac{\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}$$

Since $r^2 \leq 1$ and standard deviations σ_x, σ_y are positive.

$\therefore$ +ve sign gives the acute angle between the lines.

Hence,

$$\boxed{\tan\theta = \frac{1-r^2}{r} \cdot \frac{\sigma_x \sigma_y}{\sigma_x^2+\sigma_y^2}}$$

∴ +ve sign gives the acute angle between the lines.

Hence, $\boxed{\tan\theta = \frac{1-r^2}{r} \cdot \frac{\sigma_x \sigma_y}{\sigma_x^2+\sigma_y^2}}$

Question-16:

Discuss the regression and its importance Given the following data.

x	1	5	3	2	1	1	7	3
y	6	1	0	0	1	2	1	5

Find the regression line x on y.

⇒ Regression measures the nature and extent of correlation. Regression is the estimation or prediction of unknown values of one variable from known values of another variable. It identifies the connections between variables occuring in a dataset and determines the magnitude of these associations and their significance on outcomes.

x	y	x^2	y^2	xy
1	6	1	36	6
5	1	25	1	5
3	0	9	0	0
2	0	4	0	0
1	1	1	1	1
1	2	1	4	2
7	1	49	1	7
3	5	9	25	15
$\Sigma x = 23$	$\Sigma y = 16$	$\Sigma x^2 = 99$	$\Sigma y^2 = 68$	$\Sigma xy = 36$

Mean $\bar{x} = \frac{\Sigma x}{n}$

$= \frac{23}{8} = 2.875$

Mean $\bar{y} = \frac{\Sigma y}{n}$

$= \frac{16}{8} = 2$

∴ $\bar{x} = 2.875$ and $\bar{y} = 2$

the know, regression coefficient b_{xy} is given as

$$b_{xy} = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma y^2 - (\Sigma y)^2}$$

$$= \frac{8 \times 36 - 23 \times 16}{8 \times 68 - 16 \times 16} = \frac{\cancel{8}(36 - 23 \times 2)}{\cancel{8}(68 - 16 \times 2)}$$

$$= \frac{36 - 46}{68 - 32} = \frac{-10}{36} = -0.278$$

Now, Regression line of x on y is

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$x - 2.875 = -0.278(y - 2)$$

$$x - 2.875 = -0.278y + 0.556$$

$$x - 2.875 - 0.556 = -0.278y$$

$$x - 3.431 = -0.278y$$

Multiplying both sides by 72

$$72x - 247 = -20y$$

$$\therefore \quad 72x = -20y + 247$$

Question-17:

Write the two applications of regression.

→ Some of the use cases of regression analysis are:

- Stock Market Forecasting: Predicts price fluctuations and risk trends, helping investors optimize portfolio decisions.
- Sales Prediction: Estimates product demand accross seasons and campaigns, improving inventory and marketing planning.

- Real Estate Pricing: Calculates property value based on locality, size and economic conditions, assisting buyers and sellers.
- Healthcare Monitoring: Forecasts patient metrics such as disease progression or readmission risk for better treatment planning.
- Manufacturing Optimization: Predicts product quality and defect chances using machine parameters and sensor data.

Question-18

Ten individuals are chosen at random from a normal population of students and their marks are found to be 63, 63, 66, 67, 68, 69, 70, 70, 71, 71. In the light of these data, discuss the suggestion that mean marks of the population of students is 66.

⇒ Given, $n = 10$, $\mu = 66$

x	$x - \bar{x}$	$(x-\bar{x})^2$
63	−4.8	23.04
63	−4.8	23.04
66	−1.8	3.24
67	−0.8	0.64
68	0.2	0.04
69	1.2	1.44
70	2.2	4.84
70	2.2	4.84
71	3.2	10.24
71	3.2	10.24
$\Sigma x = 678$		$\Sigma(x-\bar{x})^2 = 81.6$

$\Sigma x = 678$

Hence, $\bar{x} = \frac{\Sigma x}{n}$

$= \frac{678}{10}$

$\bar{x} = 67.8$

$\Sigma(x-\bar{x})^2 = 81.6$

Standard deviation of population is

$$S = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}} = \sqrt{\frac{81.6}{9}} = \sqrt{9.066}$$

$$\approx 3.011$$

Null Hypothesis: $H_0: \mu = 66$, i.e; there is no significant difference between the sample and population means.

Alternative Hypothesis: $H_1: \mu \neq 66$.

Hence, we apply two-tailed test.

Test statistic: Under Ho, the test statistic is

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{67.8 - 66}{3.011/\sqrt{10}}$$

$$t = \frac{1.8 \times \sqrt{10}}{3.011} = 1.890$$

Degree of freedom $(\gamma) = n-1 = 10-1 = 9$

For, $\gamma = 9$ at a 5% level of significance, the tabulated value is, $t_{0.05} = 2.26$.

Conclusion: Since, the calculated value of t, $(|t| = 1.890)$ is less than the tabulated value of t at 5% level of significance ∴ The null hypothesis $\mu = 66$ is accepted i.e., there is no significant difference between their means.

Question-19

A random sample of 10 boys had the IQ's 70, 120, 110, 101, 88, 83, 95, 98, 107 and 100. Do these data support the assumption of a population mean I.Q of 160?

⇒ Given, $n = 10$, $\mu = 160$

Null Hypothesis: Ho: There is no significant difference in the sample mean and the population mean, i.e; $\mu = 160$.

Alternative Hypothesis: $\mu \neq 160$. (Two tailed test)

Test statistic: Under Ho, the test statistic is

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$$

x	$x-\bar{x}$	$(x-\bar{x})^2$
70	-27.2	739.84
120	22.8	519.84
110	12.8	163.84
101	3.8	14.44
88	-9.2	84.64
83	-14.2	201.64
95	-2.2	4.84
98	0.8	0.64
107	9.8	96.04
100	2.8	7.84
$\Sigma x = 972$		$\Sigma(x-\bar{x})^2 = 1833.6$

$\Sigma x = 972.$

Hence, $\bar{x} = \frac{\Sigma x}{n} = \frac{972}{10}$

$\bar{x} = 97.2$

$$S = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}} = \frac{\sqrt{1833.6}}{\sqrt{9}}$$

$$S = \frac{42.82}{3} = 14.274$$

$$t = \frac{\bar{x}-\mu}{S/\sqrt{n}} = \frac{97.2-160}{14.274/\sqrt{10}}$$

$$t = \frac{-62.8}{4.514} \approx -13.92$$

Degree of freedom $(\gamma) = n-1 = 10-1 = 9$

For $\gamma = 9$, at a 5% level of significance, the tabulated value is 2.62.

Conclusion: Since the calculated value of t $(|-13.92| = 13.92)$ is much greater than the tabulated value of t (2.26) at the 5% level of significance, the null hypothesis Ho is rejected. i.e; there is significant difference between the sample mean and population mean.

i.e; the sample could not have come from this population.

Question-20

The hight of 6 randomly chosen sailors in inches are 63, 65, 68, 69, 71 and 72. Those of 9 randomly chosen soldiers are 61, 62, 65, 66, 69, 70, 71, 72, and 73. Test whether the sailors are on the average taller than soldiers.

⇒ Given, $n_1 = 6$, $n_2 = 9$

Null Hypothesis: $H_0: \mu_1 = \mu_2$.

i.e. the mean of both the population are same

Alternative Hypothesis: $H_1: \mu_1 > \mu_2$

Calculating mean and standard deviation of first sample,

X_1	$x_1 - \overline{x_1}$	$(x_1 - \overline{x_1})^2$
63	-5	25
65	-3	9
68	0	0
69	1	1
71	3	9
72	4	16
68		60

$$\overline{x_1} = \frac{\Sigma x_1}{n_1} = \frac{408}{6} = 68$$

$$s_1^2 = \frac{\Sigma (x_1 - \overline{x_1})^2}{n_1} = \frac{60}{6}$$

Calculating mean and standard deviation for second sample,

X_2	$x_2 - \overline{x_2}$	$(x_2 - \overline{x_2})^2$
61	-6.67	44.36
62	-5.67	32.035
65	-2.67	7.0756
66	1.67	2.7556
69	1.34	1.7956
70	2.34	5.4756
71	3.34	11.1556
72	4.34	18.8356
73	5.34	28.5156
609		152.0002

$$\overline{x_2} = \frac{\Sigma x_2}{n_2} = \frac{609}{9} = 67.67$$

$$s_2^2 = \frac{\Sigma (x_2 - \overline{x_2})}{n_2} = \frac{152.0002}{9}$$

$$S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1+n_2-2} \quad \text{or} \quad S^2 = \frac{\Sigma(x_1-\bar{x}_1)^2 + \Sigma(x_2-\bar{x}_2)^2}{n_1+n_2-2}$$

$$S^2 = \frac{60+152.0002}{6+9-2} = \frac{212.0002}{13} = 16.3077$$

$$S = 4.038$$

Test statistic :

Under H_0

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{68-67.666}{4.038\sqrt{\frac{1}{6}+\frac{1}{9}}} = \frac{0.334}{4.038\sqrt{\frac{9+6}{54}}}$$

$$t = \frac{0.334}{4.038 \times 0.527} = 0.1569$$

The value of t at 5% level of significance is 1.77

Conclusion: Since $t_{calculated} < t_{0.05}$, the null hypothesis H_0 is accepted. i.e, there is no significant difference between their average. i.e, the soldiers are not on the average taller than the soldiers.

Question-24

Find the coefficient of skewness by Karl Pearson's method for the following data :

Value	6	12	18	24	30	36	42
Frequency	4	7	9	18	15	10	3

Value (x_i)	Frequency (f_i)	$x_i f_i$	x_i^2	$f_i x_i^2$
6	4	24	36	144
12	7	84	144	1008
18	9	162	324	2916
24	18	432	576	10368
30	15	450	900	13500
36	10	360	1296	12960
42	3	126	1764	5292
	66	1638		46188

$$A.M. = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{1638}{66} = 24.82$$

$$\text{Variance } (\sigma^2) = \Sigma(f_i x_i^2) - (\Sigma(f_i x_i))^2$$

$$= \frac{46188}{66} - (24.82)^2$$

$$= 699.82 - 616.0324$$

$$\sigma^2 = 83.79$$

$$S.D. = \sqrt{83.79} = 9.15$$

and Mode $= 24$

$$K_p = \frac{A.M - Mode}{S.D} = \frac{24.82 - 24}{9.15} = \frac{0.82}{9.15} = 0.09$$